

UWMamba: UnderWater Image Enhancement With State Space Model

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Abstract—Recently, state space models (SSM) with efficient design, i.e., Mamba, have shown great potential in modeling long-range dependencies with linear complexity. However, the pure SSM-based model yields sub-optimal underwater enhancement performance due to insufficient local details. Given the superiority of convolution in local perception, we propose a hybrid network, named UWMamba, which combines SSM and convolution for underwater image enhancement. We introduce a conv mamba layer (CML) as the foundation layer to combine the visual state space block (VSSB) with convolution. The convolution is used to capture local detailed features, while the VSSB is employed to capture long-range global features, which complement each other. Furthermore, considering underwater images suffer from severe and uneven degradation of spatial regions and color channels, we propose a Mamba Attention Fusion Module (MAFM), which fuses VSSB with an attention mechanism for better perception of channels and spatial regions. Extensive experiments on real-world underwater image datasets demonstrate the promising performance of our method in both objective metrics and subjective comparisons.

Index Terms—Attention mechanism, state space model, underwater image enhancement.

I. INTRODUCTION

DUE to the effects of impurities and light scattering in underwater scenes, underwater images suffer from hazing, color distortion, blurring, and other problems [1], [2]. Therefore, underwater image enhancement, aiming to improve the visual quality of images, is essential to obtain clear underwater images and investigate the underwater environment, which has a wide range of applications [3], [4] in ocean exploration, marine biology research and monitoring, and other fields.

Most of the existing data-driven based methods [5], [6], [7], [8] have achieved impressive performance on underwater image enhancement, which could be categorized into two types, the CNN-based methods [9], [10], and the Transformer-based methods [7], [11]. CNN-based methods are effective in capturing

local features, but the receptive field is limited by the kernel size, resulting in challenges in modeling global pixel dependencies. The Transformer-based methods have excellent global perception, owing to the self-attention mechanism based on image patches, but such methods require quadratic complexity of input size, resulting in a high computational burden. Recently, state space models (SSM) [12], [13] with generic and efficient design have shown great potential for modeling long-range dependencies with linear complexity. The state space model with the selective mechanism, called mamba [14], has shown superiority over Transformer on natural language processing. Meanwhile, some variants of mamba have shown remarkable performance on various vision tasks, such as image classification [15] and medical image segmentation [16], [17]. Therefore, we further explore the potential of the state space model for the underwater enhancement task in this letter.

In this work, we introduce a U-shaped pure SSM-based model inspired by [16]. However, the U-shaped pure SSM-based model yields sub-optimal performance in the underwater image enhancement task. We consider that the visual state space block (VSSB) is excellent at capturing long-range global features with linear complexity [15], but pays insufficient attention to local spatial features, which is critical for underwater image enhancement tasks. Convolutional operations excel at capturing local spatial features, which can complement the visual state space block. Therefore, we design a Conv Mamba Layer (CML) to complement SSM with local detail features. The CML combines the visual state space block with convolution. The convolution is used to capture local detailed features, while the VSSB is employed to capture long-range global features, which complement each other to extract comprehensive visual features. Different from VSSB proposed in Vision Mamba [15], we propose a Multi-Scanning 2D (MS2D) module to replace Selective Scanning 2D (SS2D) in VSSB to further improve robustness and global representation. Then, we use the conv mamba layers as the foundation layer to build a symmetric U-shaped network with an encoder and decoder for underwater image enhancement, called UWMamba. Besides, considering that underwater images suffer from severe and uneven degradation of spatial regions and color channels, we further propose a Mamba Attention Fusion Module (MAFM), which fuses VSSB with the attention mechanism, to alleviate such problem and improve the enhancement performance of UWMamba. Specifically, the Mamba Attention Fusion Module computes attention maps of the encoder features from the spatial and channel dimensions and dynamically merges the attention maps with the VSSB to improve the feature perception of channels and spatial regions. Extensive experiments on real-world underwater image datasets

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All relevant code and models can be accessed at <https://github.com/LeonSakura/UWMamba>.

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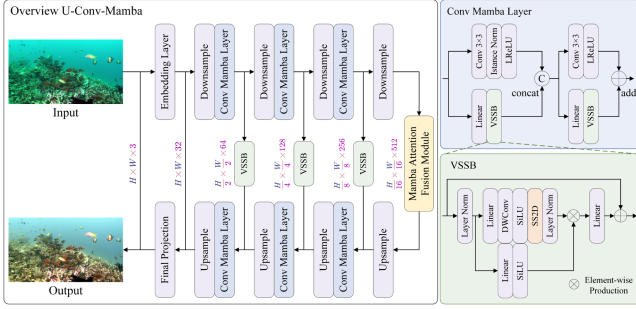


Fig. 1. The overall structure of the UWMamba. The blue box shows the detailed structure of the conv Mamba layer. The green box shows the detailed structure of the visual state space block.

demonstrate the promising performance of our method in both objective metrics and subjective comparisons. In summary, the main contributions of this paper can be summarized as follows:

- We propose a UWMamba network for underwater image enhancement, which incorporates the visual state space block for long-range global information and convolution for local perception. In VSSB, we introduce a Multi-Scanning 2D module to improve feature global representation and robustness.
- We propose a Mamba Attention Fusion Module that dynamically merges the VSSB with the attention mechanism to alleviate the uneven degradation of underwater images in spatial regions and color channels.
- We conduct extensive experiments on several real-world underwater image datasets. The results demonstrate that our UWMamba achieves superior performance compared with previous underwater image enhancement methods.

II. METHOD

This work introduces the latest advanced State Space Model (SSM), i.e., Mamba, to the convolution-based underwater image enhancement model. In this section, We give an overview of the structure of UWMamba, followed by a detailed description of the proposed Conv Mamba Layer. Besides, we propose a Mamba Attention Fusion Module to fuse the VSSB with the attention mechanism for better perception of channels and spatial regions.

A. Overall of UWMamba

The architecture of our UWMamba is shown in Fig. 1, which is inspired by Vision Mamba [15] and Unet [18]. Specifically, UWMamba is composed of a feature embedding layer, an encoder, a decoder, and skip connections. First we use the feature embedding layer based entirely on convolution to extract shallow features $F \in \mathbb{R}^{H \times W \times C}$ from the input underwater image $I_{input} \in \mathbb{R}^{H \times W \times 3}$, where H, W denote the height and width of the input image, C denotes the number of channels, which defaults to 32. Then, we use the encoder to extract deep features progressively. The encoder consists of four Conv Mamba Layers, each preceded by down-sample layers. The down-sample layer employs a maximum pooling operation to reduce the height

and width of the input features. The Conv Mamba Layer focuses on expanding the number of channels and feature extraction. The output resolution of each Conv Mamba Layer in the encoder is $\frac{H}{2} \times \frac{W}{2} \times 2C$, $\frac{H}{4} \times \frac{W}{4} \times 4C$, $\frac{H}{8} \times \frac{W}{8} \times 8C$, $\frac{H}{16} \times \frac{W}{16} \times 16C$, respectively. Similarly, the decoder is symmetrical to the encoder design, composed of three Conv Mamba Layers and a final projection layer, each preceded by up-sample layers. The up-sample layers employ the bilinear interpolation operation. Besides, unlike previous U-shape methods [7], we have not used a straightforward add operation for all skip connections. For deep features with small resolution, we employ visual State Space Block to fuse them with the features in the decoder. For shallow features with large resolution, we use a straightforward add operation to avoid introducing redundant computational complexity.

B. Conv Mamba Layer

As shown in the blue box in Fig. 1, the Conv Mamba Layer (CML) consists of two stages, and each stage splits into two branches: the local branch and the global branch. The local branch in stage 1 consists of a 3×3 convolution with stride 1, an InstanceNorm layer, and a Leaky ReLU activation function. The global branch in stage 1 consists of the linear layer and the visual state space block [16], [17]. The detailed structure of VSSB is shown in the green box in Fig. 1. Then, we concatenate the local and global features of stage 1, $f_{local}^1, f_{global}^1$, in the channel dimension, which can be expressed as:

$$z = \text{Concat}(f_{local}^1, f_{global}^1) \quad (1)$$

After that, we obtain the fusion features z of stage 1. Then, we feed the fused features z into the local and global branches in stage 2. The local branch in stage 2 consists of a 3×3 convolution with stride one and a Leaky ReLU activation function. The global branch in stage 2 remains the same as in stage 1. The local branch is used to capture local spatial detail features, while the global branch is used to capture long-range global features. Finally, we fuse global and local features of stage 2 with the addition operation to get the output features of Mamba-Conv Layer f_{MCL} :

$$f_{MCL} = \text{Add}(f_{local}^2, f_{global}^2) \quad (2)$$

Multi-Scanning 2D module is designed to better utilize 2D spatial features. As shown in Fig. 2(a), MS2D scans the input features in eight different directions, which enables the 2D features to be flattened into 1D sequences. Then, we obtain the long-range dependencies of each sequence. Finally, we use summation to merge the eight sequences and reshape the same 2D structure as the input features. Compared with SS2D [15], MS2D scans the features from multiple directions, resulting in merged features with stronger global representation and better robustness.

C. Mamba Attention Fusion Module

Considering that underwater images suffer from severe and uneven degradation of spatial regions and color channels, we further propose a Mamba Attention Fusion Module, which fuses VSSB with the attention mechanism, to alleviate such problem and improve the enhancement performance of our UWMamba.

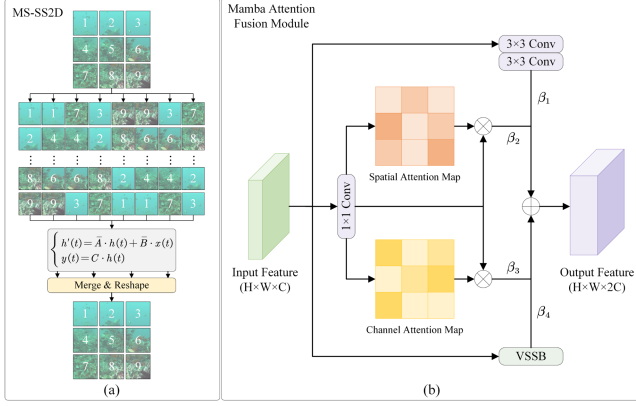


Fig. 2. (a) Structure of the MS2D. (b) Structure of MAFM. $\otimes$ denotes the matrix multiplication, and $\oplus$ denotes the summation.

The structure of the Mamba Attention Fusion Module is shown in Fig. 2(b).

After we get the output features of the encoder $f_{enc} \in \mathbb{R}^{H \times W \times C}$, a 1×1 convolution is used to transform the channel dimension of f_{enc} . Then we reshape the output features of the encoder and use matrix multiplication to compute the spatial self-attention map $Att_{sp} \in \mathbb{R}^{HW \times HW}$:

$$Att_{sp} = \text{reshape}(f'_{enc})_{HW \times 2C} \times \text{reshape}(f'_{enc})_{2C \times HW} \quad (3)$$

where $f'_{enc} \in \mathbb{R}^{H \times W \times 2C}$ are the output features of the encoder after channel transformation. Similarly, we calculate the channel self-attention map $Att_{ch} \in \mathbb{R}^{2C \times 2C}$:

$$Att_{ch} = \text{reshape}(f'_{enc})_{2C \times HW} \times \text{reshape}(f'_{enc})_{HW \times 2C} \quad (4)$$

Subsequently, we multiply f'_{enc} with the spatial self-attention map Att_{sp} and the channel self-attention map Att_{ch} , respectively, to achieve better feature representation of channels and spatial regions. Finally, we dynamically merge the processed features f_{ch} , f_{sp} with the local features f_{conv} extracted by two stacked 3×3 convolution operations and the global features f_{vssb} extracted by the VSSB. The merge feature f_{merge} can be expressed as:

$$f_{merge} = \beta_1 \cdot f_{conv} + \beta_2 \cdot f_{ch} + \beta_3 \cdot f_{sp} + \beta_4 \cdot f_{vssb} \quad (5)$$

where $\beta_1, \beta_2, \beta_3, \beta_4$ are dynamic trainable weights.

III. EXPERIMENTS

A. Implementation Details

Training Details: The proposed UWMamba is implemented by using the PyTorch framework via NVIDIA RTX4090. L1 loss and MSE loss are used as supervised functions. The total training epoch is set to 300, and the batch size is set as 16. We resize the input images to a fixed size (256×256), and employ the SGD optimizer with the initial learning rate of 0.01, the weight decay of $5e-4$, and the momentum of 0.9.

Dataset: We use the real-world UIEB dataset [19] and the LSUI dataset [7] to train and evaluate our models. The UIEB dataset contains 890 real-world underwater images with corresponding labels, and is randomly divided as UIEB-Train (800

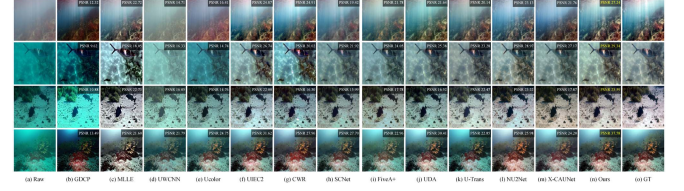


Fig. 3. Visual comparison of enhancement results sampled from UIEB-test and LSUI-test. From left to right are raw underwater images, results of GDCP [21], MLE [22], UWCNN [23], Ucolor [9], UIEC² [10], CWR [24], SCNet [25], FiveA+ [26], UDA [11], U-Trans [7], NU2Net [27], X-CAUNET [28], our UWMamba and the reference image (GT). The PSNR value is in the upper right of the images. The highest PSNR value is marked in yellow.

images) and UIEB-Test (90 images) for training and testing. The LSUI dataset comprises 4,279 real-world underwater images with corresponding labels, and is randomly divided as LSUI-Train (3879 images) and LSUI-Test (400 images) for training and testing. Besides, we further use C60 [19], U45 [2], and UCCS [20] benchmarks for non-reference quality evaluation. C60 contains 60 challenging non-reference images in UIEB. U45 comprises 45 non-reference images containing the underwater haze scene and underwater green scene. UCCS includes 300 non-reference images in tones of blue-green, green, and blue.

Evaluation Metrics: For the dataset with the reference image, we use PSNR, SSIM [29], UCIQE [30] and UIQM [31] as evaluation metrics. As for the no-reference dataset, we use UCIQE [30] and UIQM [31] as evaluation metrics.

B. Comparison With SOTA Methods

Full-reference Evaluation: We use UIEB-Test and LSUI-Test for full reference evaluation. We present the quantitative comparison and visual comparison results in Table I and Fig. 3. The UIEB row of Table I shows the quantitative results compared with different methods on UIEB. Our UWMamba outperforms all competing methods in terms of full-reference image quality evaluation metrics, whether CNN-based methods or Transformer-based methods. Besides, as shown in Table II, we compare UWMamba with UIEC² (the best performing CNN-based method) X-CAUNET (the best performing Trans-based method) in terms of FLOPs, parameters and running time for efficiency evaluation. As for running time, we calculate the average running time by testing 100 images with a size of 256×256 . Obviously, UWMamba with the lowest FLOPs and the fastest running time is the most efficient method. The LSUI row of Table I shows the quantitative results compared with different methods on LSUI. Our UWMamba achieves 30.57 PSNR and 0.926 SSIM on the LSUI dataset, which surpasses the previous state-of-the-art method. To further qualitatively analyze the visual quality, the enhanced images from different methods are shown in Fig. 3. Obviously, UWMamba can achieve more accurate color restoration with fewer color deviations. Meanwhile, UWMamba outperforms other methods in terms of texture details and contrast enhancement, clearly reflecting our superiority in enhancing underwater scene.

Non-reference Evaluation: We use C60, U45, and UCCS for non-reference evaluation. All methods are trained with

TABLE I
QUANTITATIVE COMPARISONS ON FOUR DATASETS

	Methods	Tradition-Based		CNN-Based						Trans-Based				Ours
		GDCP [21] TIP'18	MLLE [22] TIP'22	UWCNN [23] PR'20	Ucolor [9] TIP'21	UIEC ² [10] SPIC'21	CWR [24] IGARSS'22	SCNet [25] ICASSP'22	FiveA+ [26] BMVC'23	UDA [11] CG'23	U-Trans [7] TIP'23	NU2Net [27] AAAI'23	X-CAUNET [28] ICASSP'24	
UIEB	PSNR ↑	13.22	19.36	16.14	19.22	22.96	20.89	21.71	21.19	23.18	20.19	23.29	23.39	23.95
	SSIM ↑	0.697	0.830	0.707	0.859	0.874	0.798	0.870	0.861	0.908	0.783	0.901	0.902	0.910
	UCIQE ↑	0.60	0.59	0.49	0.58	0.60	0.60	0.59	0.59	0.60	0.56	0.59	0.60	0.61
	UIQM ↑	2.38	2.71	2.91	2.91	3.14	3.03	2.93	2.97	3.05	3.13	3.18	3.01	3.16
LSUI	PSNR ↑	13.48	17.52	22.42	24.74	26.73	22.12	25.09	23.92	28.85	24.78	26.48	25.20	30.57
	SSIM ↑	0.688	0.716	0.834	0.877	0.877	0.818	0.874	0.867	0.921	0.848	0.901	0.912	0.926
	UCIQE ↑	0.60	0.60	0.55	0.57	0.59	0.59	0.58	0.57	0.59	0.58	0.58	0.59	0.59
	UIQM ↑	2.54	2.33	2.92	2.77	2.89	2.89	2.97	2.85	2.90	2.94	2.99	2.74	3.00
C60	UCIQE ↑	0.57	0.56	0.51	0.54	0.53	0.55	0.53	0.54	0.56	0.54	0.55	0.55	0.57
	UIQM ↑	1.95	2.20	2.18	2.27	2.29	2.39	2.13	2.39	2.56	2.76	2.35	2.40	2.42
U45	UCIQE ↑	0.58	0.59	0.53	0.55	0.58	0.57	0.56	0.56	0.58	0.57	0.57	0.57	0.59
	UIQM ↑	2.59	2.48	3.01	2.80	3.17	2.82	3.03	3.08	3.02	3.20	3.16	2.78	3.18
UCCS	UCIQE ↑	0.55	0.57	0.49	0.52	0.56	0.55	0.53	0.53	0.56	0.53	0.54	0.55	0.56
	UIQM ↑	2.74	2.87	2.80	2.82	2.98	2.93	2.92	3.02	3.04	3.00	3.09	2.82	3.11

Full-reference evaluation metrics are PSNR, SSIM, UCIQE and UIQM. Non-reference evaluation metrics are UCIQE and UIQM. For the UIEB, all methods are trained with UIEB-train and tested on UIEB-test. For the LSUI, all methods are trained with LSUI-train and tested on LSUI-test. The highest scores are marked in red.

TABLE II
EFFICIENCY EVALUATIONS ON UIEB-TEST

Method	FLOPs(G) ↓	TIME(S) ↓	Params(M) ↓
UIEC ² [10]	26.08	0.024	0.53
X-CAUNET [28]	111.27	0.012	31.78
UWMamba	14.45	0.009	17.60

The best performance CNN model UIEC² [10] and trans model X-CAUNET [28] are compared. The highest scores are marked in red.

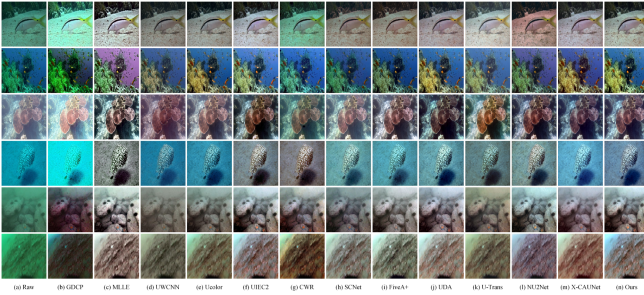


Fig. 4. Visual comparison of enhancement results from C60, U45, and UCCS. From left to right are raw underwater images, results of GDCP [21], MLLE [22], UWCNN [23], Ucolor [9], UIEC² [10], CWR [24], SCNet [25], FiveA+ [26], UDA [11], U-Trans [7], NU2Net [27], X-CAUNET [28] and UWMamba.

LSUI-Train (3879 images), then tested on C60, U45, and UCCS respectively. The quantitative comparison and visual comparison results are shown in Table I and Fig. 4. It can be observed that UWMamba achieve competitive UCIQE and UIQM scores. As for qualitative results, our method effectively removes the haze in underwater images and produces enhanced images with more realistic color. Meanwhile, our enhanced images exhibit more appropriate contrast and saturation, leading to better visual effects.

C. Ablation Study

To demonstrate the effectiveness of our UWMamba, we conduct ablation studies on different model designs using the UIEB-Test dataset. As shown in Table III, Pure Conv denotes the original U-Net [18] design with pure convolutions, and Pure SSM represents replacing all the convolution blocks in U-Net

TABLE III
EVALUATIONS OF DIFFERENT MODEL DESIGNS ON UIEB-TEST

Method	PSNR ↑	SSIM ↑	UCIQE ↑	UIQM ↑	FLOPs(G) ↓	Params(M) ↓
Pure Conv	19.37	0.834	0.57	3.30	31.04	54.75
Pure SSM	19.95	0.815	0.63	3.02	7.565	34.62
Ours w/o MAFM	23.03	0.886	0.61	3.12	13.89	15.42
Ours w/o MS2D	23.74	0.909	0.62	3.10	14.45	17.60
Ours	23.95	0.910	0.61	3.16	14.45	17.60

TABLE IV
ABLATION EXPERIMENTS OF EACH COMPONENT IN MAFM

Method	PSNR ↑	SSIM ↑	UCIQE ↑	UIQM ↑	FLOPs(G) ↓	Params(M) ↓
Only Conv	23.03	0.886	0.61	3.12	13.90	15.42
Conv + VSSB	23.42	0.886	0.60	3.13	14.34	17.14
Conv + VSSB + ATT	23.95	0.910	0.61	3.16	14.45	17.60

with visual state space blocks, following [16]. The evaluation metrics indicate that both Pure Conv and Pure SSM model designs yield sub-optimal underwater enhancement performance, highlighting the superiority of the UWMamba fusion design of convolutions and visual state space blocks. Moreover, with the MAFM and MS2D modules, UWMamba achieves superior underwater image enhancement performance with minimal additional computational complexity. As shown in Table IV, we also explore the effectiveness of each component in the MAFM.

IV. CONCLUSION

In this paper, we explore the potential of recent advanced State Space Models in the field of underwater image enhancement and propose a U-shaped network named UWMamba. To simultaneously capture local features and global information, we combine VSSB with convolution to design a conv mamba layer as the foundational layer of UWMamba. To improve robustness and global representation, we propose an MS2D module to replace the original SS2D module in VSSB. Considering the specificity of underwater scenes, we further design a Mamba Attention Fusion Module to enhance performance. Compared with other methods, our hybrid network UWMamba, based on SSM and convolution, achieves promising enhancement results. Quantitative and qualitative evaluations on multiple datasets demonstrate the superiority of UWMamba in underwater image enhancement.

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